

A Tiered Roadmap for Calculations in Modal Triplet Theory (MTT): Audited Closure and Strict Upgrades

Peter Nero

July 2026
Version 3

Abstract

We organize MTT calculations by explicit closure standard and source provenance. Embedded renormalized-Standard-Model equivalence is closed at the adopted one-shared-physical-primitive/measured-profile standard: the final audit passes 12/12 obligations using selected multi-loop transport, the 27-by-27 operator architecture, charged-flavor and CKM rows, Higgs/threshold support, and imported standard SM quantization. Independent structural results now include the native faithful Standard Model gauge group, an anomaly-free 48-state chiral carrier, a completed 96-dimensional finite triple at profile tier, an exact one-Higgs projector, and a zero-parameter 10/10 finite gauge/ghost spectrum source. These achievements do not yet derive the measured gauge and Yukawa values: the strict program remains 2/9, with U2 and U4 closed. The roadmap therefore records named source objects, scope guards, dependencies, and reproducible acceptance tests for every stronger claim.

Version 3 Revision Note

Supersedes *A Tiered Roadmap for Calculations in Modal Triplet Theory*, version 2.

Reason The former roadmap mixed obsolete benchmark calculations with open source theorems and repeatedly reopened already certified matrix, Yukawa, electroweak, and threshold work. It also preceded the native gauge-group, finite-triple, one-Higgs, and finite-spectrum results.

Resolution Version 3 locks the 12/12 embedded-renormalized-SM baseline, retires the few-TeV/CY backfit chain, and replaces repeated status targets by nine named strict-upgrade objects and acceptance tests. It incorporates the new exact finite-geometry results without promoting their still-open numerical source maps.

Retained result The current one-shared-physical-primitive/profile closure and its reproducible packets remain the operational baseline.

Remaining boundary Continuum q79 source geometry, a selected upper action, strict observable values, nonperturbative QFT, and unique-branch selection remain stronger upgrades and must not be confused with reopening the closed baseline.

1 How to read this roadmap

The word *closed* is always local to a named row and standard. For example, a finite spectrum may be derived exactly while the physical coefficient multiplying it remains profile data. Likewise, an embedding can reproduce every renormalized Standard Model observable in a common

scheme without deriving the measured source coordinates. These are different mathematical achievements, and neither should be advertised as the other.

The tiers below are therefore cumulative but not interchangeable. A higher tier may use a lower-tier theorem, but a numerical match cannot retroactively turn a calibration into a source theorem. The result matrix states both the positive result and the scope guard needed to use it correctly.

2 Closure vocabulary

Every result must carry one of the following labels:

Structural theorem. Derived from selected mathematical source data.

Conditional theorem. Derived only after all displayed assumptions are imposed.

Calibration. Parameters are chosen using the compared data.

Profile/replay. Measured coordinates are transported or embedded in a declared scheme.

Diagnostic. A consistency check whose target entered selection or fit.

Held-out prediction. The target is absent from source selection, calibration, branch choice, and uncertainty construction.

No result may move between these classes merely because its numerical residual is small.

3 Current headline theorem

The permitted headline is:

MTT closes embedded renormalized-Standard-Model equivalence at the adopted one-shared-physical-primitive/profile standard on the selected branch.

It must be accompanied by the limitation:

This is an embedding/parity result with measured profile inputs and imported standard SM quantization. It is not zero-knob derivation, unique observed-branch selection, or a derivation of perturbative quantization from MTT.

The numerical authority is the SMDR v1.3 common-scheme transport at $Q = M_t = 172.5590883453979$ GeV. Fifteen source coordinates map to eight full-SM $\overline{\text{MS}}$ rows with a positive-definite 8×8 covariance, all 36 symmetric entries, and all 15 BCT–WZH cross entries.

4 Revised tier architecture

4.1 Tier 0: conventions and provenance

Tier 0 fixes units, hypercharge normalization, renormalization scheme, scale, branch identifiers, source ownership, covariance, and code versions. It produces no physical prediction. This tier is mandatory for every numerical claim.

4.2 Tier 1: exact finite and topological structure

Tier 1 contains exact results only with their branch hypotheses. Current examples include the selected $q = 79$ modulo-448 branch, finite-gerbe/charge data, the 27-by-27 qutrit–Weyl architecture, finite Cech tables, cocycle identities, and the selected retarded orientation representative.

The native rank-one, rank-two, and rank-three bundle automorphisms give $U(1)$, $SU(2)$, and $SU(3)$ with faithful quotient $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$. On the selected three-family carrier, the chiral representation and all local and global gauge-anomaly checks close exactly. These results do not prove uniqueness over all MTT branches or automatically determine measured scalar magnitudes.

4.3 Tier 2: geometry-light conditional relations

Tier 2 now separates exact algebra from assumptions. Modal democracy implies $\sin^2 \theta_W = 3/8$ but is not selected by the current profile. Holonomy phase sums require a connection-preserving trivialization. Equal wave speeds require a common principal symbol. Curvature-mass and PPN statements require explicit mass ansatz, positivity, response norms, and physical normalization. Internal spectral gaps are not external cutoffs.

4.4 Tier 3: profile-standard superset execution

The legacy crossing-scale backfit is retired. Current Tier 3 is the audited superset embedding:

- final profile-standard audit: 12/12;
- one shared physical primitive P_{EW} and zero Higgs-specific parameters at the embedding layer;
- selected multi-loop transport and covariance;
- ten magnitude labels and physical threshold/mass coordinates retained transparently as profile data; and
- standard SM perturbative quantization imported rather than derived.

4.5 Tier 4: selected geometry and operator execution

The old fitted Calabi–Yau corner is not validated and is retired as numerical authority. The selected rank-two literal Cech–HYM execution is closed 2/2, including finite patching and the continuum tail/Wiener certificate. The internal $K_{\text{threshold}}$ response closes 10/10. Charged flavor and Higgs close at the profile standard; CKM also closes at the selected prediction-with-uncertainty criterion. Rank-three transfer, strict scalar value emission, absolute neutrino completion, and strong CP remain stronger targets.

The newer generative finite-geometry lane supplies a sharper structural result. It constructs the 48-state family-diagonal chiral carrier, the particle–antiparticle 96-dimensional finite module, its profile-valued finite Dirac operator, and the minimal neutral algebra summand required by orientability and Poincare duality. The completed triple has a single anomaly-free shared hypercharge circle. Exhaustive finite one-form execution produces three raw scalar doublets, while the selected q79/proto-spinor alignment projector isolates one exact Higgs doublet. The sector-resolved finite gauge/ghost spectra are sourced 10/10 with no continuous parameter.

This last spectrum result also proves an important limitation: after rank/adjoint normalization the selected finite rows are universal. They cannot by themselves emit the observed relative gauge couplings. The two relative gauge coordinates and the physical scale therefore remain profile inputs until a selected continuum HYM/action response supplies them.

5 Current result matrix

Area	Current status	Scope guard
Embedded renormalized SM	closed 12/12	measured profile and imported SM quantization
Native gauge group and chiral carrier	exact	selected low-energy branch; global vacuum/action selection open
Completed finite triple	closed at profile tier	physical Dirac entries are profile rows
Finite Higgs module	one-Higgs projection exact	absolute spectral moments and normalization open
Finite gauge/ghost spectra	sourced 10/10	universal normalized rows do not predict coupling ratios
27-by-27/charged operator	closed structurally	magnitudes remain profile rows at strict tier
Literal rank-two Cech–HYM	closed 2/2	no all-branch uniqueness or rank-three transfer
Internal K threshold	closed 10/10	physical threshold values remain profile rows
CKM	U4 closed	uncertainty-profile prediction, not moving-central-value identity
Neutral response	internal dimensionless 9/9	physical unit and Dirac/Majorana completion open
Strong CP	partial	selected nonzero anomaly-matching/relaxation map open
Gravity TT carrier	scoped support closed	Newton normalization and full stress response open
Quantization/QFT	restricted exact plus formal/partial	canonical binary record law and free reference net close; general Born source and nonperturbative interacting limit remain open

6 Strict-upgrade program

The baseline is not reopened by stronger goals. The nine upgrades are:

- U1. zero-primitive empirical source: open;
- U2. literal global Cech–HYM/QaSU3: closed 2/2;
- U3. official joint input likelihood: partial;
- U4. CKM prediction beyond source rows: closed at profile criterion;
- U5. absolute neutrino mass and ontology: partial;
- U6. strong-CP selection: partial;
- U7. MTT-derived quantization: partial, with a restricted canonical binary-recorder theorem;
- U8. constructive nonperturbative four-dimensional QFT: partial; and
- U9. unique observed-branch selection: partial at orientation level.

7 Next computational milestones

1. **Physical q79 endpoints.** Construct the visible and hidden holomorphic bundles in one Hull–Strominger/HYM chamber, beginning with the explicit η_9 meridian, period, and flat Deligne rows.
2. **Continuum naturality and execution.** Intertwine the local $1 + 2 + 3$ carrier with the physical q79 HYM complex, then evaluate the selected rank-102 Hessian blocks, kernel projection, and radii inequalities.
3. **Upper action.** Define one selected upper differential/action whose automorphisms, zero modes, transferred products, and finite operators reproduce the accepted lower structures.
4. **Strict observable sources.** Derive the electroweak primitive, relative gauge metrics, nine charged magnitudes, Higgs value, and physical threshold rows from that same source, or retain each explicitly as profile data.
5. **Neutral and strong-CP completion.** Supply dimensionful neutral blocks and absolute normalization; separately construct the selected flux/threshold axion-current anomaly map and test EDM constraints.
6. **Quantization and QFT.** Extend the restricted recorder theorem to all allowed contexts and construct a geometry-selected nonperturbative gauge–BRST completion with positivity and continuum control.
7. **Branch and zero-primitive synthesis.** Upgrade the retarded orientation representative to a global action/measure theorem, and attempt U1 only after its dependency rows close.

8 Reproducibility contract

Every milestone must ship raw inputs, immutable source identifiers, units, scheme/scale/loop order, branch choices, parameter ledger, covariance, executable code, deterministic outputs, unit tests, and an acceptance certificate. Held-out claims additionally require a data-separation certificate. Historical packets remain provenance and cannot override their audited successors.

9 Conclusion

The program has advanced substantially beyond the original roadmap, but not in the manner that document claimed. The strongest current achievement is a reproducible, shared-operator embedding of the renormalized Standard Model at a clearly declared profile standard, together with a native faithful gauge group, anomaly-free chiral carrier, completed finite triple, one-Higgs projector, exact 10/10 finite gauge spectra, and the HYM and CKM strict upgrades. None of these converts the measured profile coordinates into derived constants. The old Tier-3 and Calabi–Yau numerical chain is retired.

Progress toward a deeper theory is now sharply measurable: close U5, U6, U7–U9 and finally U1 with source theorems rather than fitted replacements. This roadmap is intentionally non-looping: a row changes status only when its named missing object passes its stated acceptance test.

Computational Evidence and Reproducibility

This roadmap reports the status and tier of the calculation program rather than reproving each packet. The closed and profile rows below are used directly as status evidence; the open strict-upgrade row remains an unresolved target and is not counted as closure.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper's local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A01/direct_k_higgs_row (*derived exact*).
Promoted direct K_threshold.Omega_H.lambda row.
- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.
- A01/strict_pew_row (*derived exact*).
Promoted P_EW source row at the declared one-shared-primitive standard.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

References

- [1] P. Nero, *Modal Triplet Theory: Admissibility, Encodings, and the Structure of Physical Description*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255621>
- [2] P. Nero, *Modal Triplet Theory: Foundation*, Zenodo preprint, September 2025. <https://doi.org/10.5281/zenodo.16949762>
- [3] P. Nero, *Fixed Points I–VI: Complete Coherence Spine*, Zenodo preprints, August 2025. <https://doi.org/10.5281/zenodo.16948748>
- [4] P. Nero, *The Projection–Admissibility Principle: Structural Constraints on Effective Physical Description*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255838>
- [5] P. Nero, *Closure and Inevitability in Modal Triplet Theory*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255510>
- [6] P. Nero, *Coherence Capacity as the Fundamental Resource of Effective Physics*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255905>
- [7] P. Nero, *Dynamics of Coherence Capacity: Transport, Concentration, and Exhaustion*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18256048>
- [8] P. Nero, *Modal Triplet Theory: From MTT to Quantum Mechanics*, Zenodo preprint, September 2025. <https://doi.org/10.5281/zenodo.17074246>
- [9] P. Nero, *From MTT to Quantum Field Theory*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17068816>
- [10] P. Nero, *Modal Triplet Theory: From MTT to General Relativity*, Zenodo preprint, October 2025. <https://doi.org/10.5281/zenodo.16950597>
- [11] P. Nero, *Modal Triplet Theory: From MTT to a UV-Finite, Unitary Quantum Gravity*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17077671>
- [12] P. Nero, *Measurement as Disturbance and Stabilization in Modal Triplet Theory*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17177404>
- [13] P. Nero, *Projection, Probability, and Irreversibility: Shadow Bridges Between Measurement, Black Holes, and Cosmology in Modal Triplet Theory*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18256408>
- [14] P. Nero, *Modal Fixed Points, Bell’s Beables, and the Limits of Factorization*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17076300>
- [15] P. Nero, *Temporal Bell Inequalities and Global Consistency in Modal Triplet Theory*, Zenodo preprint, August 2025. <https://doi.org/10.5281/zenodo.18208884>
- [16] P. Nero, *From Modal Triplet Theory to Indivisible Stochastic Processes: A First-Principles, Fully Rigorous Derivation*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18254862>